

Problem Set 3

Monday, September 21, 2026

7:53 PM

Consider the data: [red, blue, blue, red, green, grey]

- a. Represent these data as a matrix of one hot encoded variables
- b. Compute the sample proportions for each label

Consider the data:

$$X = [1, 3, 3, 6, 11]$$

- c. Compute and sketch the empirical CDF for this sample.

a)

	red	blue	green	grey
1	1	0	0	0
3	0	1	0	0
3	0	1	0	0
6	1	0	0	0
11	0	0	1	0
11	0	0	0	1

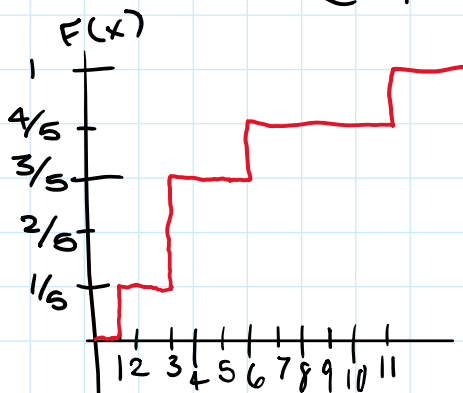
b)

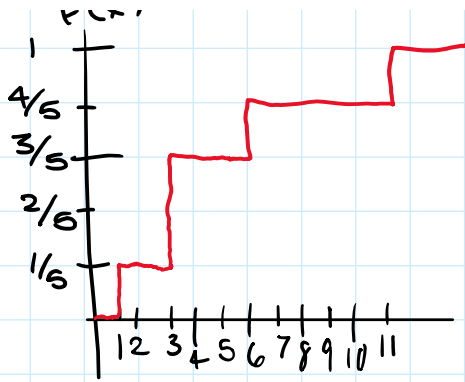
$$\hat{p}_{\text{red}} = \frac{2}{6} = \frac{1}{3} \quad \hat{p}_{\text{green}} = \frac{1}{6}$$

$$\hat{p}_{\text{blue}} = \frac{2}{6} = \frac{1}{3} \quad \hat{p}_{\text{grey}} = \frac{1}{6}$$

c) $X = [1, 3, 3, 6, 11]$ $F(x) = p[X \leq x]$

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/6 & 1 \leq x < 3 \\ 3/6 & 3 \leq x < 6 \\ 4/6 & 6 \leq x < 11 \\ 1 & 11 \leq x \end{cases}$$





Q2.

We often want to take transformations of random variables, particularly when doing feature engineering.

- a. Suppose we have a random variable X and transform it as $Y = a + bX$, with $b > 0$. The distribution of X is given by $F_X(x)$. What is the distribution of Y , $F_Y(y)$? What is the density, $f_Y(y)$?

Suppose your sample is:

$$X = [1, 3, 4]$$

- b. What is the mean of the sample?
- c. Square all the values. What the mean of X^2 ?
- d. Compute the square of the mean from part b. Does this equal the mean of the square from part c?

Non-linear transformations change the moments/statistics of the data in a non-linear way. This is important to remember.

a)

$$Y = a + bX \quad F_Y(y) = P(Y \leq y)$$

$$F_Y(y) = P(a + bX \leq y)$$

$$F_Y(y) = P\left(X \leq \frac{y-a}{b}\right)$$

$$F_Y(y) = F_X\left(\frac{y-a}{b}\right)$$

$$f_Y(y) = \frac{d}{dy} F_X\left(\frac{y-a}{b}\right)$$

$$= \frac{1}{b} f_X\left(\frac{y-a}{b}\right)$$

b) $\bar{X} = \frac{1+3+4}{3} = \left\lfloor \frac{8}{3} \right\rfloor$

$$b) \bar{x} = \frac{1+3+4}{3} = \boxed{\frac{8}{3}}$$

$$c) \overline{x^2} = \frac{1+9+16}{3} = \boxed{\frac{26}{3}}$$

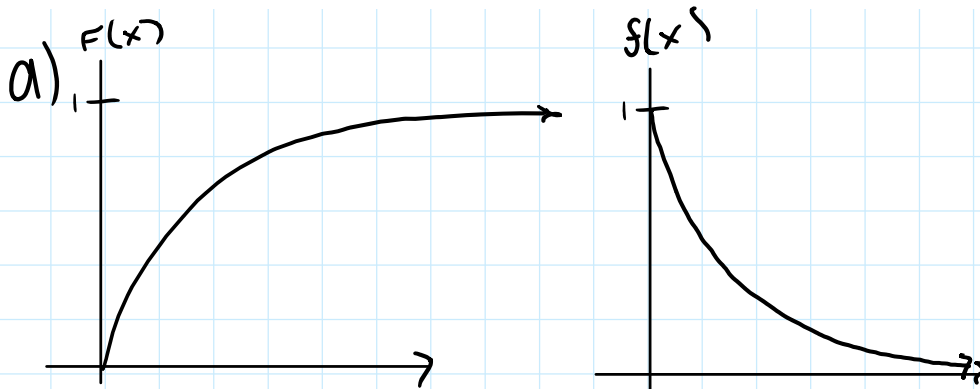
$$d) (\bar{x})^2 = \boxed{\frac{64}{9} \neq \frac{26}{3}}$$

Q3.

The exponential distribution is

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

- a. Sketch a graph of the exponential distribution and density.
- b. What is the probability that $X \geq 3$? What is the probability that $X \leq 35$?
- c. Suppose we transform X by $Y = 1 + 3X$. What is the distribution of Y , $F_Y(y) = pr[Y \leq y]$? Provide a formula and sketch a graph in comparison to F_X . What is the probability that Y is less than 10?



$$b) P(X \geq 3) = 1 - F_X(3) = 1 - (1 - e^{-3}) \approx \boxed{0.05}$$

$$P(X \leq 35) = F_X(35) = 1 - e^{-35} \approx 1$$

my calculator refuses to show actual value b/c e^{-35} is so tiny

$$c) Y = 1 + 3X$$

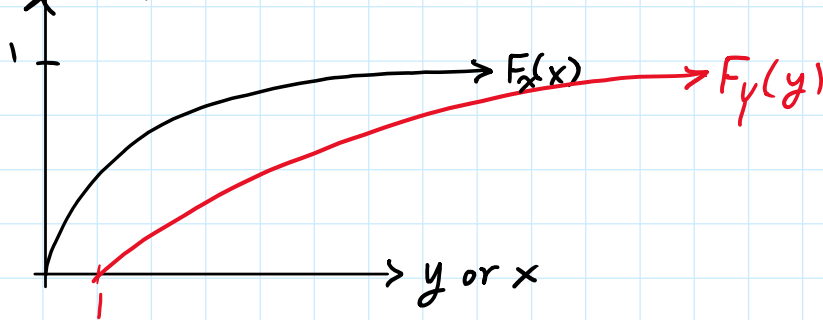
$$P(Y < y) = P(1 + 3X < y)$$

$$= P(X < \frac{y-1}{3})$$

$$F_Y(y) = F_X\left(\frac{y-1}{3}\right)$$

$$F_Y(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-(y-1)/3} & y \geq 1 \end{cases}$$

$F_Y(y)$ or $F_X(x)$



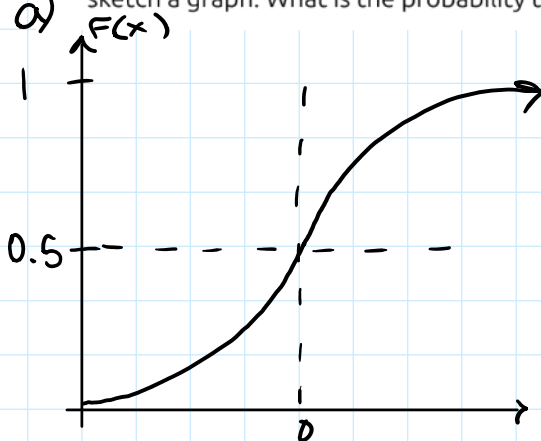
$$P(Y < 10) = F_Y(10) = 1 - e^{-(10-1)/3} = 0.9502$$

Q4

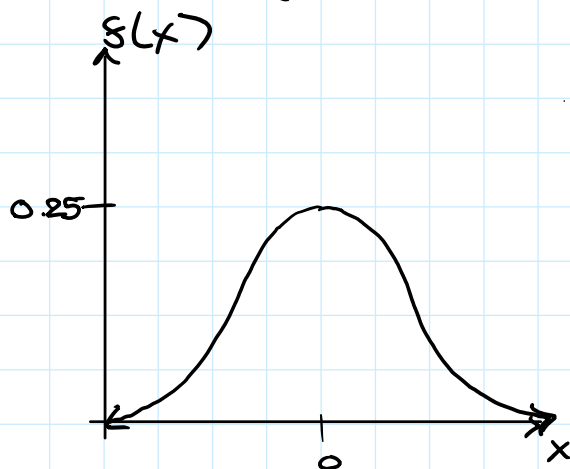
$$F(x) = \frac{1}{1 + e^{-x}}$$

- a. Sketch a graph of the logistic distribution and density.
- b. What is the probability that $X \geq .8$? What is the probability that $X \leq .3$?
- c. Suppose we transform X by $Y = 2X - 1$. What is the distribution of Y ? Provide a formula and sketch a graph. What is the probability that Y is less than 0?

a)



$$f(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$



b)

$$P(X \geq 0.8) = 1 - F(X \geq 0.8)$$

b)

$$P(X \geq 0.8) = 1 - F(X \geq 0.8)$$

$$= 1 - \frac{1}{1 + e^{-0.8}} = \boxed{0.31}$$

$$P(X \leq 0.3) = F(X \leq 0.3)$$

$$= \frac{1}{1 + e^{-0.3}} = \boxed{0.57}$$

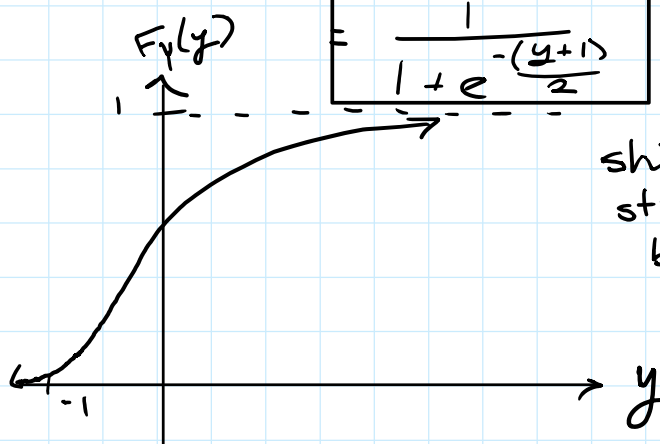
c)

$$F_Y(y) = P(Y < y) = P(2X - 1 < y)$$

$$= P(X < \frac{y+1}{2})$$

$$= F_X\left(\frac{y+1}{2}\right)$$

$$= \frac{1}{1 + e^{-\frac{(y+1)}{2}}}$$



shifts -1, &
stretches horizontally
by 2

$$P(Y < 0) = \frac{1}{1 + e^{-\frac{(0+1)}{2}}} = \boxed{0.62}$$

Q5.

The **median** is the value x for which the probability that X is above or below x is .5, or $F(\text{median}) = \frac{1}{2}$.

- a. What is the median of the exponential distribution?
- b. What is the median of the logistic distribution?

The **quantile function** is the inverse of the distribution function. The CDF answers the question, "What fraction u of the time is X below x ?" ($F(x) = u$) and the quantile function answers the question, "For what value x is X below x with probability u ?" ($F^{-1}(u) = x$)

- c. What is the quantile function of the exponential distribution?
- d. What is the quantile function of the logistic distribution?

$$a) \quad \frac{1}{2} = 1 - e^{-x} = \quad b) \quad \frac{1}{2} = \frac{1}{1 + e^{-x}}$$

- d. What is the quantile function of the logistic distribution?

$$a) \quad \frac{1}{2} = 1 - e^{-x} =$$

$$\Rightarrow \frac{1}{2} = e^{-x}$$

$$\ln\left(\frac{1}{2}\right) = -x$$

$$x = -\ln\left(\frac{1}{2}\right)$$

$$x = \ln(2)$$

$$b) \quad \frac{1}{2} = \frac{1}{1 + e^{-x}}$$

$$\Rightarrow 1 + e^{-x} = 2$$

$$e^{-x} = 1$$

$$x = 0$$

$$c) \quad u = F_X(x) = 1 - e^{-x}$$

$$u = 1 - e^{-x}$$

$$\Rightarrow x = -\ln(1 - u)$$

$$F^{-1}(u) = -\ln(1 - u)$$

$$d) \quad u = \frac{1}{1 + e^{-x}}$$

$$1 + e^{-x} = \frac{1}{u}$$

$$e^{-x} = \frac{1 - u}{u}$$

$$x = -\ln\left(\frac{1 - u}{u}\right)$$

$$F^{-1}(u) = -\ln\left(\frac{1 - u}{u}\right)$$

Q6.

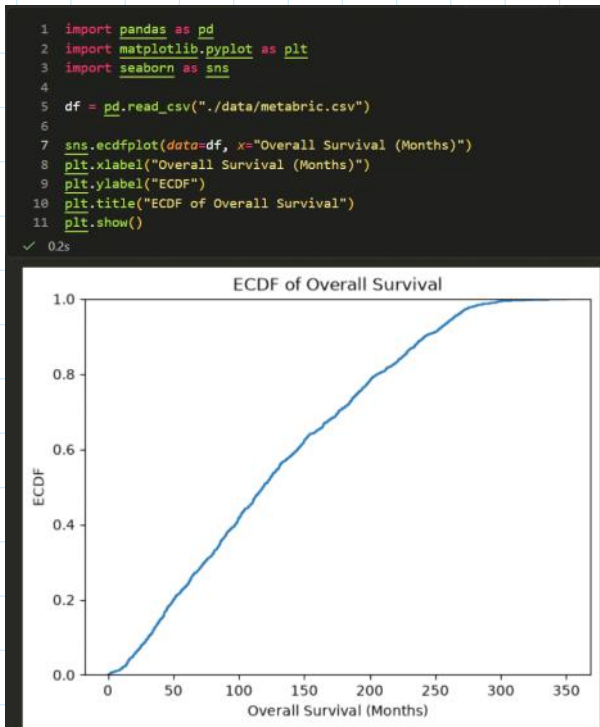
Load `./data/metabric.csv`.

- a. Make an ECDF plot of **Overall Survival (Months)**.
- b. Make an ECDF plot of **Overall Survival (Months)**, hued by **Chemotherapy**. Conditional on a patient receiving Chemotherapy, should we predict they will live a longer or shorter amount of time? Explain your answer clearly.
- c. Is chemotherapy an effective treatment? Explain your answer clearly.

a)

b)

a)



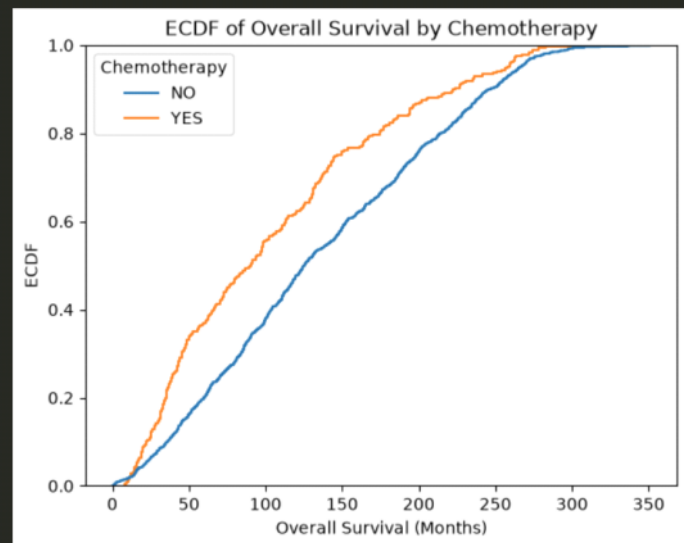
b)

```

1 sns.ecdfplot(data=df, x="Overall Survival (Months)", hue="Chemotherapy")
2 plt.xlabel("Overall Survival (Months)")
3 plt.ylabel("ECDF")
4 plt.title("ECDF of Overall Survival by Chemotherapy")
5 plt.show()

```

✓ 0.3s



We can predict that someone who is receiving chemotherapy is going to live less time because the chemotherapy ECDF line is further to left, meaning that those receiving chemotherapy live less time. This is because people that are receiving chemotherapy are more likely to have a more severe type/stage of cancer that requires such an aggressive treatment, thereby meaning that they are more likely to live a shorter amount of time.

- C) We cannot determine from this ECDF alone whether chemotherapy is an effective treatment. The ECDF shows an association between chemotherapy status and overall survival, but it does not necessarily mean that it is an ineffective treatment. Patients who received chemotherapy may have differed from patients who did not receive chemotherapy in important ways, such as disease severity.